

EE63050/AI2100/AI5100 Deep Learning, Fall 2024

Indian Institute of Technology Hyderabad

Homework 2, Mathematical Preliminaries, Assigned 05.09.2024, Due 11:59 pm on 13.09.2024

Do. Or do not. There is no try. – Jedi Master Yoda

Instructions:

- It is **strongly recommended** that you work on your homework on an *individual* basis. If you have any questions or concerns, feel free to talk to the instructor or the TAs.
- **The functions implemented for HW1 can be reused here.**
- Use `matplotlib` where specified - <https://matplotlib.org/tutorials/introductory/images.html>.
- Do not use other built-in functions that directly solve a problem.
- Please turn in Python Notebooks with the following notation for the file name: `your-roll-number-hw2.ipynb`.
- Do not turn in images. Please use the same names for images in your code as in the database. The TAs will use these images to test your code.

Problem Set:

1. **Distance between PDFs:** In this question you will explore the other “distances” between PDFs discussed in class. To verify the implementation of these distances, use the normalized histogram of the stereo image pair (`left.png`, `right.png`) used in the previous assignment.
 - (a) **Cross Entropy (CE):** The cross entropy between two PDFs (PMFs) p and q is given by: $H(p, q) = H(p) + D(p||q)$ where $H(p)$ is the entropy of p and $D(p||q)$ is the KL divergence between p and q . Write a function that accepts two PDFs (PMFs) p, q and outputs the CE between them.
 - i. Verify your function using the stereo image normalized histogram pair. (1)
 - ii. As with the KL divergence problem, choose a fixed PMF $p \sim \text{Bern}(r)$. Choose another PMF $q \sim \text{Bern}(s)$ where s can be varied. Plot $H(p, q)$ as a function of s . From the plot, does minimizing $H(p, q)$ give us matched PMFs? (1)
 - (b) **Jensen Shannon (JS) Divergence:** The definition of JS divergence between two PDFs p and q is given by: $J(p, q) = 0.5 * (D(p||m) + D(q||m))$ where $m = \frac{p+q}{2}$ and $D(p||q)$ is the KL divergence between p and q . Write a function that accepts two PDFs (PMFs) p, q and outputs the JS divergence between them. Verify that the $JS(p, q)$ is symmetric indeed while $D(p||q)$ is not. Again, use the normalized histograms of the stereo image pair. (1)
 - (c) **Wasserstein Distance:** The Wasserstein-1 distance between two PDFs r and s is given by: $W_1(r, s) = \inf_{\pi \in \Pi(r, s)} \mathbb{E}_{(x, y) \sim \pi} |x - y|$. The set $\Pi(r, s)$ is composed of all bivariate joint PDFs whose marginals equal r and s . Given a tuple $(p_{(X, Y)}, r_X, s_Y)$ of a joint histogram $p_{(X, Y)}$, and marginals r_X, s_Y , write a function that accepts this tuple and checks if $p_{X, Y} \in \Pi(r, s)$. Verify your function with a positive example and a negative example. (2)
2. **Visualizing Data Using t-SNE:**
 - (a) Read the [t-SNE paper](#) and answer the following questions. *Do not reproduce text from the paper verbatim in your answers.*
 - i. What is the crowding problem? (1)
 - ii. How does the choice of the Student t-distribution in the low dimensional embedding space help address the crowding problem? (1)
 - iii. What other important changes have been made in t-SNE relative to SNE? (1)

- (b) In this problem, implement Algorithm 1 from the paper, albeit in a simplified setting as described in the following. (5)
- Generate two clusters of points from a ten-dimensional multivariate Gaussian (MVG) distribution $\mathcal{N}(\mu, 0.01 \cdot I)$ where I is the ten-dimensional identity matrix.
 - Use $\mu_1 = \mathbf{1}$ for one cluster and $\mu_2 = 10 \cdot \mathbf{1}$ for the other (where $\mathbf{1}$ is the ten-dimensional vector of ones).
 - Generate 10 points from each cluster for a total of 20 points to form the set $\mathcal{X}$.
 - Choose the dimension of the embedding to be two.
 - Choose $T = 50$.
 - Experiment with different choices for η and $\alpha(t)$. For simplicity, let $\alpha(t)$ not change with iterations.
 - Use your knowledge of how $\mathcal{X}$ was generated for choices of σ_i (as opposed to finding them using the user-defined *Perplexity*).
 - Plot the points in $\mathcal{Y}$ at the beginning and at the end of 50 iterations. Print your observations from the plots.
 - Find and print $D(P||Q)$ at the beginning and at the end of 50 iterations. Print your observations from these values.
 - The YouTube video by the first author Laurens van der Maaten can be found [here](#).
- (c) Now, experiment with the built-in t-SNE utility in `matplotlib`. Choose four different perplexity values (between 5 and 50) and generate t-SNE plots for these choices. How does perplexity affect the plots? (2)